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A General Method for Calculating Error Probabilities over Fading Channels

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Abstract — This paper presents a general method for calculating the average error rates and outage performance of a broad class of coherent, differentially coherent and noncoherent communication systems with/without diversity reception in a myriad of fading environments. Unlike the moment generating function (MGF) technique, the proposed characteristic function (CHF) method based on Parseval's theorem enables us to unify the average error rate analysis of different modulation formats and all commonly used predetection diversity techniques (i.e., maximal-ratio combining (MR), equal-gain combining (EG), selection diversity (SD), switched diversity (SW) and hybrid diversity systems) in a single common framework. The CHF method also lends itself to the averaging of the conditional error probability (CEP) involving the complementary incomplete Gamma function and the confluent hypergeometric function over fading amplitudes, which heretofore resisted to a simple form. As an aside, we show previous results as special instances of our unified framework.

I. INTRODUCTION

In communications systems analyses over wireless channels, we frequently encounter the task of averaging the conditional error probability (CEP) over fading amplitudes or the received signal power. The CEPs for binary and M-ary modulation formats with coherent, differentially coherent or noncoherent detection schemes are usually in the form of an exponential function $\exp(.)$, complementary error function $\mathrm{erfc}(.)$ (or Gaussian probability integral $Q(.)$), a combination of $\mathrm{erfc}(.)$ and $\mathrm{erfc}^2(.)$, complementary incomplete Gamma function $\Gamma(.,.)$, confluent hypergeometric series ${}_1F_1(.;.;.)$, Marcum-Q function $Q(.,.)$, and generalized Marcum-Q function $Q_M(.,.)$ (see [1]-[4] and all the related references found therein). One of the most commonly used technique to accomplish this task in the past has been the probability density function (PDF) method, viz.,

$$P_S(\varepsilon, L) = \int_0^\infty P_S(\varepsilon|\gamma)p_\gamma(\gamma, L)d\gamma \qquad (1)$$

where $P_S(\varepsilon|\gamma)$ denotes the symbol error probability in AWGN channel conditioned on the signal-to-noise ratio (SNR) at the combiner output, L is the diversity order, and $p_\gamma(\gamma, L)$ corresponds to the PDF of combiner output SNR in a specified fading environment. This approach has been used by many authors over the past five decades to analyze various single channel reception systems performance over fading channels. Sometimes, it is more convenient to evaluate (1) if the CEP and the PDF are expressed in terms of the combiner output envelope $\vartheta = \sqrt{\gamma}$ [5],

$$P_S(\varepsilon, L) = \int_0^\infty P_S(\varepsilon|\vartheta)p_\vartheta(\vartheta, L)d\vartheta \qquad (2)$$

Diversity systems and multichannel signaling introduces a new wrinkle to the problem on hand because we now need to determine the PDF of a sum of random variables (RVs) in case of MR or EG diversity systems. There are several variations of the PDF method, which can be categorized depending as to how the PDF is obtained. In the most traditional form, the evaluation of (1) or (2) will require an L-fold convolution integral. In most cases, the Fourier transform of the PDF is more readily available than the PDF itself. Thus, the PDF can be expressed as

$$p_x(x, L) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \psi_x(\omega, L)e^{-j\omega x}d\omega \qquad (3)$$

where $\psi_x(\omega, L)$ is the CHF of random variable x at the combiner output and $x \in \{\gamma, \vartheta\}$. This CHF is also related to the MGF (i.e., Laplace transform of the PDF) via relationship $\psi_x(\omega, L) = \langle e^{j\omega x}\rangle = \phi_x(-j\omega, L)$ or alternatively, $\phi_x(s, L) = \langle e^{-sx}\rangle = \psi_x(js, L)$. For some special cases (e.g., sum of exponential RVs), the inverse Fourier transform (FT) (3) can be evaluated in a closed-form, and therefore closed-form solution for the PDF is available in these situations. However, it is difficult (if not impossible) to get closed-form PDFs for all common fading environments, especially for the diversity systems. In this case, one may resort to an *approximate* PDF which can be easily determined using a Fourier series technique [6],

$$p_x(x, L) \approx \frac{2}{T}\sum_{\substack{n=1\\ n\ \mathrm{odd}}}^{\infty}[\psi_x(n\omega_0, L)e^{-jn\omega_0 x} + \psi_x(-n\omega_0, L)e^{jn\omega_0 x}] \qquad (4)$$

where $\omega_0 = 2\pi/T$ and the coefficient T is selected such that $Pr(x > T) \le \varepsilon$, and ε can be set to a very small value. A simple method to bound the systematic errors arise in the use (4) is presented in [7]. This variation of the PDF method has been widely used in the analysis of equal-gain diversity receivers and the co-channel interference analysis. Whereas in [8]-[9], the authors have derived PDF of the

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composite phase of the fading signal and noise, and subsequently determined the average symbol error rate for M-ary signals from its cumulative distribution function (CDF).

More recently, some authors have used MGF techniques to analyze the performance of a broad class of modulation formats in different fading environments [1-3, 10-11]. The key idea is to express the CEP in a desirable exponential form (which can be partitioned into a product form) so that the averaging can be easily performed with the knowledge of the Laplace or Fourier transform of the PDF. However, there are several limitations to this approach: (a) it fails to work in the analysis of EG diversity receivers because of the cross-product terms in the CEP except for the cases involving only erf(.) or $Q(.)$ functions[1]; (b) it fails to work if CEP cannot be expressed in a desirable exponential form (e.g., trigonometric integral for the complementary incomplete Gamma function given in [12] is not in a "desired" form)[2]; (c) it fails to yield an exact analytical expression for binary DPSK or binary orthogonal signalling with post-detection EG combining due to the limitation of trigonometric integral representation for the generalized Marcum-Q function; and (d) it fails to analyze the performance of MFSK with post-detection EG combining (square-law combining).

Recognizing that the product integral in (1) can be easily transformed into the frequency domain with the aid of Parseval's theorem, and the Fourier transform (FT) of the PDF is the CHF, we immediately obtain a simple expression for computing (1) in the frequency domain. However, in this case we also need the knowledge of FT of the CEP, which turns out to be easily computed. Therefore, we have *reformulated the task of finding a desirable exponential form for the CEP (required for the MGF method) to simply computing the FT of the CEP, which is a simpler task and generally works for all forms of the CEPs*! Note the implications of our results: (a) first of all, we have developed a *general method for calculating the average error probability performance with single and multiple channel reception in a single common framework* (MGF method cannot facilitate the analysis of EG in a common framework); (b) more importantly, the *CHF method overcomes all the limitations of the MGF method* highlighted above; and (c) if the CEP can be expressed in a desirable exponential form, then it can be shown (using Cauchy's integral formula) that the solution from the CHF method reduces to the familiar expression obtained via MGF method. Therefore, *the results obtained from the CHF method encapsulates those of the MGF approach.*

1. A desirable exponential form for $Q(.)$ suitable for EG receiver analysis is given in [5].
2. A desirable exponential trigonometric integral for $\Gamma(\alpha, \beta)$ when $0 < \alpha < 1$ is obtained in [13].

II. FREQUENCY DOMAIN ANALYSIS

By transforming (1) and/or (2) into the frequency domain, we find

$$\begin{aligned} P_S(\varepsilon, L) &= \frac{1}{2\pi}\int_{-\infty}^{\infty} G_\gamma(\omega)\psi_\gamma(\omega, L)d\omega \\ &= \frac{1}{2\pi}\int_{-\infty}^{\infty} G_\vartheta(\omega)\psi_\vartheta(\omega, L)d\omega \end{aligned} \tag{5}$$

since $\gamma \geq 0$, $\vartheta \geq 0$, $G_\gamma(\omega) = \mathrm{FT}[P_S(\varepsilon \mid \gamma)]$ and $G_\vartheta(\omega) = \mathrm{FT}[P_S(\varepsilon \mid \vartheta)]$. Expressing $G_\gamma(.)$ and $\psi_\gamma(.,.)$ or $G_\vartheta(.)$ and $\psi_\vartheta(.,.)$ in the polar form and then after simplification, (5) reduces to

$$P_S(\varepsilon, L) = \frac{1}{\pi}\int_0^{\pi/2} \frac{\mathrm{Real}\{G_x(\tan\theta)\psi_x(\tan\theta, L)\}}{\cos^2\theta}d\theta \tag{6}$$

where $x \in \{\gamma, \vartheta\}$. Notice that (6) is an *exact* finite-range integral which is suitable for numerical integration (i.e., the integrand is well behaved even at $\theta = 0$). It is also worth noting that one may also arrive at (5) by substituting (3) into (1) and/or (2) and then rearranging the order of integration. Similarly, by substituting (4) into (1), we get an *approximate* solution for $P_S(\varepsilon, L)$:

$$\begin{aligned} P_S(\varepsilon, L) &\approx \frac{2}{T}\sum_{\substack{n=1 \\ n\ \mathrm{odd}}}^{\infty} G_x(n\omega_0)\psi_x(n\omega_0, L) + G_x(-n\omega_0)\psi_x(-n\omega_0, L) \\ &\approx \frac{2\omega_0}{\pi}\sum_{\substack{n=1 \\ n\ \mathrm{odd}}}^{\infty} \mathrm{Real}\{G_x(n\omega_0)\psi_x(n\omega_0, L)\} \end{aligned} \tag{7}$$

Interestingly, this turns out to be a trapezoidal rule approximation of (6).

Since the MGF or CHF of the fading statistics is readily available for both single and multichannel reception, the outage probability can be easily calculated by invoking the Fourier inversion formula,

$$\begin{aligned} P_{out} &= \frac{1}{2} - \frac{1}{\pi}\int_0^{\infty} \frac{1}{\omega}\mathrm{Imag}[\psi_x(\omega, L)\exp(-j\omega x^*)]d\omega \\ &= \frac{1}{2} - \frac{2}{\pi}\int_0^{\pi/2} \frac{1}{\sin 2\theta}\mathrm{Imag}[\phi_x(-j\tan\theta, L)\exp(-jx^*\tan\theta)]d\theta \end{aligned} \tag{8}$$

since it gives the relationship between the CDF ($F_x(.,.), x \in \{\gamma, \vartheta\}$) and CHF (or MGF). Now using trapezoidal rule approximation, (8) can be re-stated as

$$P_{out} \approx \frac{1}{2} - \frac{2}{\pi}\sum_{\substack{n=1 \\ n\ \mathrm{odd}}}^{\infty} \frac{1}{n}\mathrm{Imag}\{\psi_x(n\omega_0, L)\exp(-jn\omega_0 x^*)\} \tag{9}$$

where ω_0 is defined as in (4).

A more rigorous treatment of the derivations of (6) and (9) from the first principle, along with six selected applications highlighting the benefits of our CHF approach as a unified mathematical tool for performance analysis of digital communications is presented in [4]. Next, we discuss some examples of these applications.

III. APPLICATIONS

In this subsection, we briefly highlight the applications of our technique to analyze a broad class of modulation formats in a variety of fading environments. The CHFs for

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Table 1. PDFs and CHFs of the fading signal amplitude and the SNR for several fading channel models.

Channel Model	PDF $f(.)$ and CHF $\psi(.)$ of fading amplitude (envelope) $\upsilon_l = \alpha_l\sqrt{E_s/N_0}$ and SNR $\gamma_l = \upsilon_l^2$ of the l-th branch
Rayleigh	$f_{\upsilon_l}(z) = \frac{2z}{\bar{\gamma}_l}\exp\left(\frac{-z^2}{\bar{\gamma}_l}\right)\ z \geq 0$ $f_{\gamma_l}(z) = \frac{1}{\bar{\gamma}_l}\exp\left(\frac{-z}{\bar{\gamma}_l}\right)$ where $\bar{\gamma}_l = \frac{E_s}{N_0}E[\alpha_l^2]$ = average SNR per symbol $\psi_{\upsilon_l}(\omega) = {}_1F_1\left(1;\frac{1}{2};\frac{-\bar{\gamma}_l\omega^2}{4}\right) + j\omega\sqrt{\frac{\pi\bar{\gamma}_l}{4}}\exp\left(\frac{-\bar{\gamma}_l\omega^2}{4}\right)$ $\psi_{\gamma_l}(\omega) = \frac{1}{1-j\omega\bar{\gamma}_l}$
Rician	$f_{\upsilon_l}(z) = \frac{2(1+K_l)z}{\bar{\gamma}_l}\exp\left(-K_l - \frac{(1+K_l)z^2}{\bar{\gamma}_l}\right)$ $\times I_0\left(2z\sqrt{\frac{K_l(K_l+1)}{\bar{\gamma}_l}}\right),\ z \geq 0$ $f_{\gamma_l}(z) = \frac{1+K_l}{\bar{\gamma}_l}\exp\left(-K_l - \frac{(1+K_l)z}{\bar{\gamma}_l}\right)I_0\left(2\sqrt{\frac{K_l(K_l+1)z}{\bar{\gamma}_l}}\right)$ where Rice factor $K_l \geq 0$ $\psi_{\upsilon_l}(\omega) = \exp(-K_l)\left[\sum_{i=0}^{\infty}\frac{K_l^i}{i!}\,{}_1F_1\left(i+1;\frac{1}{2};\frac{-\bar{\gamma}_l\omega^2}{4(1+K_l)}\right)\right.$ $\left.+j\omega\sqrt{\frac{\bar{\gamma}_l}{1+K_l}}\sum_{i=0}^{\infty}\frac{\Gamma(i+3/2)K_l^i}{(i!)^2}\,{}_1F_1\left(i+\frac{3}{2};\frac{3}{2};\frac{-\bar{\gamma}_l\omega^2}{4(1+K_l)}\right)\right]$ $\psi_{\gamma_l}(\omega) = \frac{1+K_l}{1+K_l-j\omega\bar{\gamma}_l}\exp\left(\frac{j\omega K_l\bar{\gamma}_l}{1+K_l-j\omega\bar{\gamma}_l}\right)$
Nakagami-q ($-1 \leq b \leq 1$)	$f_{\upsilon_l}(z) = \frac{2z}{\bar{\gamma}_l\sqrt{1-b_l^2}}\exp\left(\frac{-z^2}{[1-b_l^2]\bar{\gamma}_l}\right)I_0\left(\frac{b_l z^2}{[1-b_l^2]\bar{\gamma}_l}\right),\ z \geq 0$ $f_{\gamma_l}(z) = \frac{1}{\bar{\gamma}_l\sqrt{1-b^2}}\exp\left(\frac{-z}{[1-b^2]\bar{\gamma}_l}\right)I_0\left(\frac{bz}{[1-b^2]\bar{\gamma}_l}\right)$ where $b_l = (1-q_l^2)/(1+q_l^2)$ and the fading parameter $0 \leq q_l \leq \infty$ $\psi_{\upsilon_l}(\omega) = \sqrt{1-b_l^2}\sum_{i=0}^{\infty}\frac{(b_l/2)^{2i}\Gamma(2i+1)}{(i!)^2}$ $\times\,{}_1F_1\left(2i+1;\frac{1}{2};\frac{-\bar{\gamma}_l[1-b_l^2]\omega^2}{4}\right) + j\omega\sqrt{\bar{\gamma}_l}\,[1-b_l^2]$ $\times\sum_{i=0}^{\infty}\frac{(b_l/2)^{2i}\Gamma(2i+3/2)}{(i!)^2}\,{}_1F_1\left(2i+\frac{3}{2};\frac{3}{2};\frac{-\bar{\gamma}_l[1-b_l^2]\omega^2}{4}\right)$ $\psi_{\gamma_l}(\omega) = \frac{1}{\sqrt{[1-j\omega\bar{\gamma}_l(1+b_l)][1-j\omega\bar{\gamma}_l(1-b_l)]}}$
Nakagami-m	$f_{\upsilon_l}(z) = \frac{2}{\Gamma(m_l)}\left(\frac{m_l}{\bar{\gamma}_l}\right)^{m_l} z^{2m_l-1}\exp\left(\frac{-m_l z^2}{\bar{\gamma}_l}\right)\ z \geq 0$ $f_{\gamma_l}(z) = \frac{1}{\Gamma(m_l)}\left(\frac{m_l}{\bar{\gamma}_l}\right)^{m_l} z^{m_l-1}\exp\left(\frac{-m_l z}{\bar{\gamma}_l}\right)$ where Nakagami-m fading parameter $m_l \geq 0.5$ $\psi_{\upsilon_l}(\omega) = {}_1F_1\left(m_l;\frac{1}{2};\frac{-\bar{\gamma}_l\omega^2}{4m_l}\right)$ $+j\omega\frac{\Gamma(m_l+1/2)}{\Gamma(m_l)}\sqrt{\frac{\bar{\gamma}_l}{m_l}}\,{}_1F_1\left(m_l+\frac{1}{2};\frac{3}{2};\frac{-\bar{\gamma}_l\omega^2}{4m_l}\right)$ $\psi_{\gamma_l}(\omega) = \left(\frac{m_l}{m_l-j\omega\bar{\gamma}_l}\right)^{m_l}$

both fading amplitude and SNR for several commonly used fading channel models typical of terrestrial and satellite communications systems are tabulated in Table 1. In Table 2 and Table 3, we derive the FTs for various forms (mathematical functions) of CEPs normally encountered in the communications systems analysis.

Table 2. Fourier transform of CEPs

$P_S(\varepsilon\|\gamma)$	$G_\gamma(\omega) = \mathrm{FT}[P_S(\varepsilon\|\gamma)] = \int_0^\infty P_S(\varepsilon\|\gamma)e^{-j\omega\gamma}d\gamma$
1. $a\exp(-b\gamma)$	$G_\gamma(\omega) = \frac{a}{b+j\omega}$
2. $a\,\mathrm{erfc}(\sqrt{b\gamma})$	$G_\gamma(\omega) = \frac{a}{j\omega}\left[1-\sqrt{\frac{b}{b+j\omega}}\right]$
3. $a\,\mathrm{erfc}(\sqrt{b\gamma})-c\,\mathrm{erfc}^2(\sqrt{b\gamma})$	$G_\gamma(\omega) = \frac{a}{j\omega}\left[1-\sqrt{\frac{b}{b+j\omega}}\right]$ $-\frac{c}{j\omega}\left[1-\frac{4}{\pi}\frac{\tan^{-1}(\sqrt{1+j\omega/b})}{\sqrt{1+j\omega/b}}\right]$
4. $\frac{\Gamma(a,b\gamma)}{2\Gamma(a)}$	$G_\gamma(\omega) = \frac{1}{2j\omega}\left[1-\left(\frac{b}{b+j\omega}\right)^a\right]$
5. $\gamma^{a-1}\exp(-b\gamma)$	$G_\gamma(\omega) = \frac{\Gamma(a)}{[b^2+\omega^2]^{a/2}}\exp[-ja\tan^{-1}(\omega/b)]$
6. $e^{-a\gamma}\,{}_1F_1(b;c;d\gamma),\ a>d$	$G_\gamma(\omega) = \frac{1}{(a+j\omega)}\,{}_2F_1\left(b,1;c;\frac{d}{a+j\omega}\right)$
7. $\int_0^\eta a(\theta)\exp[-\gamma b(\theta)]d\theta$	$G_\gamma(\omega) = \int_0^\eta\frac{a(\theta)}{b(\theta)+j\omega}d\theta$
8. $Q(a\sqrt{\gamma},b\sqrt{\gamma})-\frac{\eta}{1+\eta}\times$ $I_0(ab\gamma)\exp\left[-\gamma\left(\frac{a^2+b^2}{2}\right)\right]$ where $b>a$	$G_\gamma(\omega) = \frac{4b^2}{\kappa_2(\kappa_2-\kappa_1)}-\frac{2\eta}{(1+\eta)\kappa_2}$ where $\kappa_1 = 2j\omega+a^2-b^2$ $\kappa_2 = \sqrt{[(a+b)^2+2j\omega][(a-b)^2+2j\omega]}$
9. $Q_M(a\sqrt{\gamma},b\sqrt{\gamma})$ where $b>a$	$G_\gamma(\omega) = \frac{4b^2}{\kappa_2(\kappa_2-\kappa_1)}+\sum_{n=1}^{M-1}\frac{2}{\kappa_2}\left(\frac{2b^2}{\kappa_2+\kappa_3}\right)^n$ where $\kappa_3 = 2j\omega+a^2+b^2$

A. Single Channel Reception

Using (6) in conjunction with the entries listed in Table 1 and Table 2 or Table 3, we can get exact analytical expressions for computing the ASER of a wide range of coherent, differentially coherent and noncoherent communication systems over generalized fading channels. It should be pointed out, however, that it appears much simpler to obtain the final result from FT identities tabulated in Table 2 (compared to Table 3). Notice also that the application of (6) does not require the CEP to be in an exponential form (as in the MGF technique) and thus this approach lends itself to a simple approach for computing the average error rates for a broader class of modulation formats (e.g., see entries 4-6 in Table 2).

Let us now consider the computation of ASER for an arbitrary two dimensional signal constellations over wireless channels. Using geometric relations, Craig intelligently obtained the CEP as a weighted sum of probabilities for all decision subregions of every possible signal point:

$$P_S(\varepsilon|\gamma) = \sum_{k=1}^{S}\frac{W_k}{2\pi}\int_0^{\eta_k}\exp\left[\frac{-\gamma a_k\sin^2(b_k)}{\sin^2(\theta+b_k)}\right]d\theta \qquad (10)$$

where S is the total number of signal points or decision subregions, W_k is the *a priori* probability that the k-th symbol is transmitted, η_k, a_k and b_k are parameters relating to decision subregion k and they are independent of γ. The corresponding ASER can be readily shown as

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$$
\begin{aligned}
P_S(\varepsilon, L) &= \sum_{k=1}^{S} \frac{W_k}{2\pi} \int_0^{\eta_k} \left[\frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{\psi_\gamma(\omega)}{\dfrac{a_k \sin^2(b_k)}{\sin^2(\theta + b_k)} + j\omega} d\omega \right] d\theta \\
&= \sum_{k=1}^{S} \frac{W_k}{2\pi} \int_0^{\eta_k} \left[\psi_\gamma(\omega) \Big|_{\omega = \frac{j a_k \sin^2(b_k)}{\sin^2(\theta + b_k)}} \right] d\theta \\
&= \sum_{k=1}^{S} \frac{W_k}{2\pi} \int_0^{\eta_k} \phi_\gamma\left(\frac{a_k \sin^2(b_k)}{\sin^2(\theta + b_k)} \right) d\theta
\end{aligned}
\tag{11}
$$

by substituting entry 7 of Table 2 into (5) and then applying Cauchy's theorem (after interchanging the order of integration). Therefore, it is evident that if the CEP can be expressed in a "desirable" exponential form, then the solution obtained from the CHF method collapses to the MGF approach[3]. Aside from this, the CHF approach may also yield an alternative exact integral expression for the ASER when the CEP has both exponential and non-exponential representations [4].

In order to avoid repetition of derivations for each different modulation formats, we conclude this subsection by noting that the ASER expressions for binary and M-ary linearly modulated signals (MPSK, MQAM, star-QAM), MDPSK, MFSK and $\pi/4$-DQPSK in a variety of fading environments can be derived in a similar fashion. The approach taken to achieve these results make use of (6), CHFs listed in Table 1, FT identities listed in Table 2, and Cauchy's theorem when an exponential representation for the CEP is available.

B. Multichannel Reception

A unique feature of our CHF approach is that it facilitates a unified analysis of a wide range of modulation formats in a variety of fading environments for all commonly used diversity combining techniques in a single common framework. This particular task was heretofore resisted to a simple solution, due to the difficulty arising from EG receiver analysis. From (6) and our previous examples, it is apparent that only the knowledge of $\psi_\gamma(., L)$ or $\psi_\vartheta(., L)$ (whichever is applicable) for each type of diversity combining scheme is further required in order to evaluate the error performance of binary and M-ary modulation formats with diversity reception. Fortunately, these CHFs can be determined quite easily for many cases of practical interest, including hybrid diversity systems.

B.1 Independent Fading

B.1.1 Maximal-Ratio Combining

From the definition of CHF, it is easy to show that

$$
\psi_\gamma^{(MR)}(\omega, L) = E\left[\exp\left(j\omega \sum_{k=1}^{L} \gamma_k \right) \right] = \prod_{k=1}^{L} \psi_{\gamma_k}(\omega) \tag{12}
$$

where $E[x]$ denotes the expectation (statistical average) of random variable x, and $\psi_{\gamma_k}(.)$ corresponds to the CHF of SNR of the k-th diversity branch. Substituting (12) into (6), and utilizing the CHF of SNR (for single channel reception) listed in Table 1 and FT of CEPs listed in Table 2, we get an exact analytical expression for the ASER with MR diversity over generalized fading channels.

3. The trigonometric integral in (11) can be expressed in closed-form in Rayleigh and Nakagami-m channels with positive integer fading severity index [14].

Table 3. Fourier transform of CEPs

$P_S(\varepsilon \mid \vartheta)$	$G_\vartheta(\omega) = \mathrm{FT}[P_S(\varepsilon \mid \vartheta)] = \int_0^\infty P_S(\varepsilon \mid \vartheta) e^{-j\omega\vartheta} d\vartheta$
1. $a \exp[-b\vartheta^2]$	$G_\vartheta(\omega) = \frac{a}{2b}\left[\sqrt{\pi b}\exp\left(\frac{-\omega^2}{4b}\right) - j\omega\, {}_1F_1\left(1;\frac{3}{2};\frac{-\omega^2}{4b}\right)\right]$
2. $a\,\mathrm{erfc}(\sqrt{b}\,\vartheta)$	$G_\vartheta(\omega) = \frac{a}{\sqrt{\pi b}}\, {}_1F_1\left(1;\frac{3}{2};\frac{-\omega^2}{4b}\right) - \frac{ja}{\omega}\left[1 - \exp\left(\frac{-\omega^2}{4b}\right)\right]$
3. $a\,\mathrm{erfc}(\sqrt{b}\,\vartheta) - c\,\mathrm{erfc}^2(\sqrt{b}\,\vartheta)$	$G_\vartheta(\omega) = \frac{(a-2c)}{\sqrt{\pi b}}\, {}_1F_1\left(1;\frac{3}{2};\frac{-\omega^2}{4b}\right) + \frac{2c}{\sqrt{2\pi b}}\, {}_1F_1\left(1;\frac{3}{2};\frac{-\omega^2}{8b}\right)\exp\left(\frac{-\omega^2}{8b}\right) + \frac{j(c-a)}{\omega}\left[1 - \exp\left(\frac{-\omega^2}{4b}\right)\right] - \frac{j\omega c}{2b\pi}\left[{}_1F_1\left(1;\frac{3}{2};\frac{-\omega^2}{8b}\right)\right]^2$
4. $\frac{\Gamma[a, b\vartheta^2]}{2\Gamma(a)}$	$G_\vartheta(\omega) = \frac{\Gamma(2a)}{[\Gamma(a)]^2 2^{2a}}\sqrt{\frac{\pi}{b}}\, {}_1F_1\left(\frac{1}{2}+a;\frac{3}{2};\frac{-\omega^2}{4b}\right) - \frac{j}{\omega}\left[\frac{1}{2} - \frac{\Gamma(2a)\sqrt{\pi}}{\Gamma(a)\Gamma(a+1/2)2^{2a}}\, {}_1F_1\left(a;\frac{1}{2};\frac{-\omega^2}{4b}\right)\right]$
5. $\int_0^\eta a(\theta)\exp[-\vartheta^2 b(\theta)]d\theta$	$G_\vartheta(\omega) = \frac{1}{2}\int_0^\eta \frac{a(\theta)}{b(\theta)}\left\{\sqrt{\pi b(\theta)}\exp\left[\frac{-\omega^2}{4b(\theta)}\right] - j\omega\, {}_1F_1\left(1;\frac{3}{2};\frac{-\omega^2}{4b(\theta)}\right)\right\}d\theta$
6. $\int_0^\eta a\exp\left[\frac{-b\vartheta^2}{\sin^2(\theta+c)}\right]d\theta$	$G_\vartheta(\omega) = \frac{a\sqrt{\pi}\cos c}{2\sqrt{b}}\exp\left(\frac{-\omega^2}{4b}\sin^2 c\right) \times {}_1F_1\left(1;\frac{3}{2};\frac{-\omega^2\cos^2 c}{4b}\right) - \frac{a\sqrt{\pi}\cos(\eta+c)}{2\sqrt{b}} \times \exp\left[\frac{-\omega^2}{4b}\sin^2(\eta+c)\right]{}_1F_1\left(1;\frac{3}{2};\frac{-\omega^2\cos^2(\eta+c)}{4b}\right) - \frac{j\omega a}{2}\int_0^\eta \frac{\sin^2(\theta+c)}{b}\, {}_1F_1\left(1;\frac{3}{2};\frac{-\omega^2\sin^2(\theta+c)}{4b}\right)d\theta$
7. $\int_0^\eta a\exp\left[\frac{-b\vartheta^2}{\cos^2(\theta+c)}\right]d\theta$	$G_\vartheta(\omega) = \frac{-a\sqrt{\pi}\sin c}{2\sqrt{b}}\, {}_1F_1\left(1;\frac{3}{2};\frac{-\omega^2\sin^2 c}{4b}\right) \times \exp\left(\frac{-\omega^2}{4b}\cos^2 c\right) + {}_1F_1\left(1;\frac{3}{2};\frac{-\omega^2\sin^2(\eta+c)}{4b}\right) \times \frac{a\sqrt{\pi}\sin(\eta+c)}{2\sqrt{b}}\exp\left[\frac{-\omega^2}{4b}\cos^2(\eta+c)\right] - \frac{j\omega a}{2}\int_0^\eta \frac{\cos^2(\theta+c)}{b}\, {}_1F_1\left(1;\frac{3}{2};\frac{-\omega^2\cos^2(\theta+c)}{4b}\right)d\theta$

B.1.2 Equal-Gain Combining

Since the envelope of the EG combiner output is $\vartheta = \frac{1}{\sqrt{L}}\sum_{k=1}^{L}\alpha_k$, its CHF is given by

$$
\psi_\vartheta^{(EG)}(\omega, L) = E\left[\exp\left(\frac{j\omega}{\sqrt{L}}\sum_{k=1}^{L}\alpha_k\right)\right] = \prod_{k=1}^{L}\psi_{\alpha_k}(\omega/\sqrt{L}) \tag{13}
$$

where $\psi_{\alpha_k}(.)$ is the CHF of the fading envelope of the k-th diversity branch. Similar to the MR case, the exact ASER with EG diversity can be computed quite easily by substituting (13) into (6), except now using the FT of CEPs listed in Table 3.

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B.1.3 Selection Diversity

The CDF of SNR at SD combiner output is simply the product of CDFs of SNR from the individual diversity branches. Then exploiting the FT of a derivative property, we get

$$\begin{aligned}\psi_\gamma^{(SD)}(\omega, L) &= -j\omega\int_0^\infty e^{j\omega\gamma}\prod_{k=1}^{L} F_{\gamma_k}(\gamma)d\gamma - \prod_{k=1}^{L} F_{\gamma_k}(0) \\ &= \sum_{i=1}^{N} w_i \prod_{k=1}^{L} F_{\gamma_k}(j\chi_i/\omega) + R_N\end{aligned} \tag{14}$$

where $F_{\gamma_k}(.)$ denotes the CDF of SNR of the k-th diversity branch (can be expressed in closed-form for Rayleigh, Rician and Nakagami-m channels [15]), χ_i and w_i are the i-th abscissa and weight respectively of the N-th order Laguerre polynomial, and R_N is the remainder term. Substituting (14) into (6) and using $G_\gamma(.)$ tabulated in Table 2, we obtain a simple expression for characterizing the SD receiver performance in different fading environments.

B.1.4 Switched Diversity

The derivation of CHF of γ in a switched diversity scheme is slightly more involved compared to the ideal SD or MR. Using a discrete-time model, it can be shown that the CHF of SNR at the output of a dual-branch switch-and-stay diversity combiner is given by [15, Eq. (10)]

$$\psi_\gamma^{(SW)}(\omega, L) = \sum_{k=1, l\neq k}^{2} \frac{F_{\gamma_l}(\xi)}{F_{\gamma_k}(\xi) + F_{\gamma_l}(\xi)}[\lambda_k(\xi, \omega) + \psi_{\gamma_l}(\omega)F_{\gamma_k}(\xi)] \tag{15}$$

where ξ denotes the fixed switching threshold and $\lambda_k(\xi, \omega) = \int_\xi^\infty \exp(j\omega\gamma)f_{\gamma_k}(\gamma)d\gamma$ is the marginal CHF of SNR of the k-th diversity branch (which can be expressed in closed-form for Rayleigh, Rician and Nakagami-m channels [15]). Interestingly, the final ASER expression for SW and SD systems is identical to the MR case, with the exception that the expression for $\psi_\gamma^{(MR)}(., L)$ is now replaced with $\psi_\gamma^{(SW)}(., L)$ and $\psi_\gamma^{(SD)}(., L)$, respectively.

B.2 Correlated Fading

It is further noted that (6) also applies to correlated fading cases. For instance, the error rate performance of an L-branch MR receiver in arbitrarily correlated Rician, Nakagami-m or Rayleigh fading channels can be readily evaluated by substituting the corresponding CHFs presented in the Appendix B of [4] into (6).

In addition, analytical expressions for calculating the ASER for dual-diversity MR, SD and SW systems over correlated Nakagami-m channels with arbitrary parameters were also derived in [4]. Using a new trigonometric integral representation for $Q_M(\alpha, \beta)$ derived in [16] (which holds for any real order M), it is also straightforward to show that the CDF (and thus the outage probability) and the PDF of an SD output SNR in bivariate Nakagami-m fading can be computed efficiently via an integral expression with finite integration limits. Further applications of the CHF approach have been discussed in depth in [4], including the development of an exact finite-range integral expression for computing the ASER of M-ary orthogonal signalling with post-detection EG (square-law combining) over generalized fading channels (i.e., by invoking the Parseval theorem twice to transform two product integrals into frequency domain).

IV. CONCLUSIONS

A simple, direct method of computing the average bit or symbol error rates with/without diversity in a *single common framework* is outlined. The FT of CEP and CHF of signal amplitude and/or SNR are used to unify the performance evaluation of the average error rates over generalized fading channel for both independent and correlated fading cases. We also point out that the outage probability for single and multichannel reception cases can be evaluated directly using only the CHF of combiner output statistic.

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